

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY HYDERABAD
B.TECH. MINOR IN CYBER SECURITY

COURSE STRUCTURE AND SYLLABUS

(Applicable for the batches admitted from the academic year 2022-2023)

V SEMESTER

R22

Course Code	Title of the Course	L	T	P/D	CH	C
22MC1CY301	Principles of Information Security	3	0	0	3	3
22MC2CY301	Principles of Information Security Laboratory	0	0	3	3	1.5
Total		3	0	3	6	4.5

VI SEMESTER

R22

Course Code	Title of the Course	L	T	P/D	CH	C
22MC1CY302	Foundations of Cyber Security	3	1	0	4	4
Total		3	1	0	4	4

VII SEMESTER

R22

Course Code	Title of the Course	L	T	P/D	CH	C
22MC1CY401	Ethical Hacking	3	0	0	3	3
22MC2CY401	Ethical Hacking Laboratory	0	0	3	3	1.5
Total		3	0	3	6	4.5

VIII SEMESTER

R22

Course Code	Title of the Course	L	T	P/D	CH	C
22MC1CY402	Incident Response Management	3	0	0	3	3
22MP1CY401	Mini – Project	0	0	4	4	2
Total		3	0	4	7	5

L – Lecture T – Tutorial P – Practical D – Drawing
 C – Credits SE – Sessional Examination CA – Class Assessment
 SEE – Semester End Examination D-D – Day to Day Evaluation
 CP – Course Project PE – Practical Examination

CH – Contact Hours/Week
 ELA – Experiential Learning Assessment
 LR – Lab Record

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. Minor (CYS) V Semester

(22MC1CY301) PRINCIPLES OF INFORMATION SECURITY

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To understand computer networks, security attacks, services, and mechanisms
- To describe various cryptosystems - symmetric key cryptography and public key cryptography
- To apply authentication services, mechanisms, and secure hash algorithms
- To be familiar with the concepts of email security, IDS, SSL, TLS, viruses, and Firewalls

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Analyse the concepts of computer networks, cryptography, information security and its applications

CO-2: Build a security model to prevent and detect the attacks using various mechanisms

CO-3: Examine the authenticity of the messages, communicate securely and investigate non-repudiation

CO-4: Apply the concepts of SSL, TLS, firewalls and establish trusted systems

UNIT-I:

Introduction to Computer Networks, Network hardware, Network software, OSI and TCP/IP Reference models, Security attacks, Security Services and Mechanisms.

UNIT-II:

Integer Arithmetic, Modular Arithmetic, Traditional Symmetric Key Ciphers, Data Encryption Standard (DES), Advanced Encryption Standard (AES).

UNIT-III:

Mathematics of Cryptography: Primes, Primality Testing, Factorization, Chinese Remainder Theorem, Asymmetric Cryptography: Introduction, RSA Cryptosystem, Rabin Cryptosystem, Elliptic Curve Cryptosystem.

UNIT-IV:

Message Integrity and Message Authentication: Message Authentication Code (MAC), SHA-512 - Digital Signatures.

UNIT-V:

Security at the Application Layer: PGP and S/MIME. Security at Transport Layer: SSL and TLS. -Principles of IDS, Virus, Firewalls, Virus Counter measures – Firewall Design Principles – Trusted Systems.

TEXT BOOKS:

1. Computer Networks, Andrew S. Tanenbaum, David. J. Wetherall, 5th Edition, Pearson Education/PHI

2. Cryptography & Network Security, Behrouz A. Forouzan, Special Indian Edition, Tata McGraw-Hill

REFERENCES:

1. Network Security Essentials (Applications and Standards), William Stallings, Pearson Education
2. Cryptography and Network Security – Principles and Practices, William Stallings, 4th Edition, Prentice Hall of India, 2005
3. Security in Computing, Charles B. Pfleeger, Shari Lawrence Pfleeger, 3rd Edition, Pearson Education, 2003

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. Minor (CYS) V Semester

(22MC2CY301) PRINCIPLES OF INFORMATION SECURITY LABORATORY

TEACHING SCHEME		
L	T/P	C
0	3	1.5

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE OBJECTIVES:

- To apply algorithms used for data encryption and decryption
- To demonstrate IDS tools
- To apply algorithms used for message Integrity and authentication of data
- To understand various protocols for information security to protect against the threats in the networks

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Implement various encryption and decryption algorithms

CO-2: Identify the emerging areas in information security

CO-3: Interpret good security practices for information security

CO-4: Demonstrate the process of data protection from various threats

LIST OF EXPERIMENTS:

1. Write a program to perform encryption and decryption using the following substitution ciphers.
 - a. Caesar cipher
 - b. Play fair cipher
 - c. Hill Cipher
2. Write a program to implement the DES algorithm.
3. Write a program to implement RSA algorithm.
4. Calculate the message digest of a text using the SHA-1 algorithm.
5. Working with sniffers for monitoring network communication (Wireshark).
6. Configuring S/MIME for email communication.
7. Using Snort, perform real time traffic analysis and packet logging.

TEXT BOOKS:

1. Cryptography and Network Security, William Stallings, 3rd Edition, Pearson Education
2. Applied Cryptography, Bruce Schneier, John Wiley, 1996

REFERENCES:

1. Cryptography and Network Security, Behrouz A. Forouzan, McGraw-Hill, 2008
2. Security in Computing, Charles B. Pfleeger, Shari Lawrence Pfleeger, 3rd Edition, Pearson Education, 2003

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. Minor (CYS) VI Semester

(22MC1CY302) FOUNDATIONS OF CYBER SECURITY

TEACHING SCHEME		
L	T/P	C
3	1	4

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To summarize various types of cyber-attacks and cyber-crimes
- To understand cyber laws and the concepts of digital forensics
- To discuss safety measures for the protection of mobile and wireless devices
- To learn the organizational security implications and threats
- To study the impact of data privacy attacks on various domains

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Identify the need of cyber security and various types of attacks

CO-2: Understand national and international regulations of cyber security and cyber forensics

CO-3: Interpret the security challenges related to mobile and wireless devices

CO-4: Analyze the security and privacy implications of an organization

CO-5: Examine the data privacy concepts and cybercrime in different domains

UNIT-I:

Introduction to Cyber Security: Basic Cyber Security Concepts, layers of security, Vulnerability, threat, Harmful acts, Internet Governance – Challenges and Constraints, Computer Criminals, CIA Triad, Assets and Threat, motive of attackers, active attacks, passive attacks, Software attacks, hardware attacks, Cyber Threats-Cyber Warfare, Cyber Crime, Cyber terrorism, Cyber Espionage, etc., Comprehensive Cyber Security Policy.

UNIT-II:

Cyberspace and the Law & Cyber Forensics: Introduction, Cyber Security Regulations, Roles of International Law. The INDIAN Cyberspace, National Cyber Security Policy, Historical background of Cyber forensics, Digital Forensics Science, The Need for Computer Forensics, Cyber Forensics and Digital evidence, Forensics Analysis of Email, Digital Forensics Lifecycle, Forensics Investigation, Challenges in Computer Forensics

UNIT-III:

Cybercrime: Mobile and Wireless Devices: Introduction, Proliferation of Mobile and Wireless Devices, Trends in Mobility, Credit card Frauds in Mobile and Wireless Computing Era, Security Challenges Posed by Mobile Devices, Registry Settings for Mobile Devices, Authentication service Security, Attacks on Mobile/Cell Phones, Organizational security Policies and Measures in Mobile Computing Era, Laptops.

UNIT-IV:

Cyber Security: Organizational Implications: Introduction, cost of cybercrimes and IPR issues, web threats for organizations, security and privacy implications, social media marketing: security risks and perils for organizations, social computing and the associated challenges for organizations

UNIT-V:

Privacy Issues: Basic Data Privacy Concepts: Fundamental Concepts, Data Privacy Attacks, Data linking and profiling, privacy policies and their specifications, privacy policy languages, privacy in different domains- medical, financial, etc. Cybercrime: Examples and Mini-Cases Examples: Official Website of Maharashtra Government Hacked, Indian Banks Lose Millions of Rupees, Parliament Attack, Pune City Police Bust Nigerian Racket, e-mail spoofing instances. Mini Cases: The Indian Case of online Gambling, An Indian Case of Intellectual Property Crime, Financial Frauds in Cyber Domain.

TEXT BOOKS:

1. Cyber Security Understanding Cyber Crimes, Nina Godbole and Sunit Belpure, Computer Forensics and Legal Perspectives, Wiley
2. Computer and Cyber Security: Principles, Algorithm, Applications, and Perspectives, B. B. Gupta, D. P. Agrawal, Haoxiang Wang, CRC Press, 2018

REFERENCES:

1. Cyber Security Essentials, James Graham, Richard Howard and Ryan Otson, CRC Press
2. Introduction to Cyber Security, Chwan-Hwa (John) Wu, J. David Irwin, CRC Press, T&F Group

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. Minor (CYS) VII Semester

(22MC1CY401) ETHICAL HACKING

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Operating Systems, Computer Networks, Network Security & Cryptography

COURSE OBJECTIVES:

- To understand the basic concepts of ethical hacking
- To learn the enumeration and system hacking process
- To understand the threats of Trojans, backdoors, and sniffers
- To learn session hijacking and its various techniques, as well as DoS, DDoS, and botnet attacks
- To learn to defend against database attacks and understand social engineering techniques

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand ethical hacking and footprinting process

CO-2: Analyze various enumeration and system hacking methods

CO-3: Identify, analyze, and counter cybersecurity risks posed by Trojans, backdoors, and sniffers

CO-4: Apply various techniques and methods used in session hijacking, Dos, DDOS and Botnets

CO-5: Detect and mitigate SQL injection vulnerabilities and implement effective policies to counter social engineering risks

UNIT-I:

Ethical Hacking Basics: Definition of Ethical Hacking, Importance of Ethical Hacking, White Hat, Black Hat, and Grey Hat Hackers, The Ethical Hacker's Process, Active vs. Passive Reconnaissance, OSINT (Open-Source Intelligence) Tools, Google Dorking and Search Engine Footprinting.

UNIT-II:

Scanning and Enumeration: Network Scanning Techniques, Port Scanning (Nmap, Netcat), Vulnerability Scanning (Nessus, OpenVAS), Windows Enumeration, Windows Security, NetBIOS and LDAP Enumeration, SNMP Enumeration, Linux/UNIX Enumeration, NTP Enumeration, SMTP Enumeration.

UNIT-III:

System Hacking and Sniffers: Password Cracking Techniques, Privilege Escalation (Linux & Windows), Spyware, Keyloggers, and Rootkits. Passive Sniffing, Active Sniffing, Address Resolution Protocol, ARP Poisoning and Flooding, Tools for Sniffing, Wireshark, Other Sniffing Tools.

UNIT-IV:

Session Hijacking and Denial of Service: Transport Layer Hijacking, Predict the Sequence Number, Take One of the Parties Offline, Take Control of the Session, Application Layer Hijacking, Session Sniffing, Predictable Session Token ID, Man-in-the-Middle Attacks, Types of DoS, Bandwidth Attacks, SYN Flood Attacks, Program and Application Attacks, Distributed Denial of Service, DDoS Tools.

UNIT-V:

Social Engineering and Phishing: Types of Social Engineering, Person-to-Person Social Engineering, Computer-Based Social Engineering, Reverse Social Engineering, Policies and Procedures. Phishing, Spear Phishing, and Whaling Attacks, Spoofing Attacks (Email, Caller ID, SMS).

Database Attacks Database Hacking, Identifying SQL Servers, SQL Injection Vulnerabilities, SQL Injection Hacking Tools.

TEXT BOOKS:

1. Certified Ethical Hacker (CEH) Cert guide, Michael Gregg, 3rd Edition, Pearson, 2019
2. Hands-On Ethical Hacking and Network Defense, Michael T. Simpson, Kent Backman, James E. Corley, 3rd Edition, Cengage Learning, 2017

REFERENCES:

1. Ethical Hacking and Countermeasures Attack Phases, EC-Council, Cengage Learning.
2. CEH: Official Certified Ethical Hacker Review Guide, Kimberly Graves, Wiley, 2007

ONLINE RESOURCES:

1. Ethical Hacking, SWAYAM-NPTEL

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. Minor (CYS) VII Semester

(22MC2CY401) ETHICAL HACKING LABORATORY

TEACHING SCHEME		
L	T/P	C
0	3	1.5

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE OBJECTIVES:

- To introduce the methodologies framework tools of ethical hacking to get awareness in enhancing the security
- To get knowledge on various attacks and their detection
- To apply Investigate web-based hacking methods
- To perform penetration testing and understanding tools

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand modern tools in Ethical Hacking

CO-2: Demonstrate the knowledge of Ethical Hacking Techniques

CO-3: Analyze Ethical Hacking methods in different situations

CO-4: Evaluate security in various web applications

LIST OF EXPERIMENTS / PROGRAMS:

WEEK 1:

Installation of virtual box and kali Linux.

WEEK 2:

Implementation of Finding IP Geo location information, IP Geo location lookup tools. Demonstration using IP location .net, dnsstuff.com and zoomeye.org.

WEEK 3:

Demonstration Session of Competitive Intelligence websites www.alexas.com, thesearchmonitor.com, Study nercraft.com, earth.google.com, shodan.io, censys.io, wappalyzer.com, httrack.com and visualping.io.

WEEK 4:

Study of packetbuilder2.0-colasoft.com, megaping tool-angryip scanner, mac address lookup uic.win/en/mac.

WEEK 5:

Demonstration Session of Advanced -IP-scanner.com, NETbios enumeration, Enumeration tool: SNMP, LDAP, NTP, and DNS.

WEEK 6:

Setup and study Vulnerability tools: Nessus professional, GFI languard, Openvas, Retina cs, Qualys free scan, Nitko, Microsoft base line security analyzer, Automated vulnerable detection system.

WEEK 7:

Study of RIG exploit kit, command shell trojan, remote access trojan, backdoor trojans.

WEEK 8:

Implement PING Method and DNS Method Sniffing: Wireshark, ARP poisoning, DNS poisoning. Hacking wireless networks and Denial of service attacks.

WEEK 9:

Setup an environment to demonstrate Social Engineering attacks, Active online attack: Hash injection attack, LLMNR/NBT-NS POISONING, Tools to create Rainbow tables: RIGEN and WINRTGEN softlookup.com, NTLM authentication processes, password Staking, Tools to execute the password hacking PW dump file.

WEEK 10:

Demonstration Session of Password cracking tool: LOPHTCRACK and OPHCRACK, how passwords are stored in windows SAM? system hacking concepts, passview tool usb, windows password recovery tools.

WEEK 11:

Study of Rootkits: Horse pill and Grayfish, How Rootkit works, SIREFEE and NECURS, Detecting Rootkits, Cyber Renaissance-Steganography.

WEEK 12:

Implement SQL injection-Testfire.net, SQL Injection techniques, File Injection Attacks, LDAP Injection Attacks, cross-site request forgery Attack (CSRF).

TEXTBOOKS:

1. The Basics of Hacking and Penetration Testing, Patrick Engebretson, Elsevier, 2013
2. Network Security and Ethical Hacking, Rajat Khare, Luniver Press, 2006
3. Introduction to Computer Networks and Cyber Security, Wu and J. D. Irwin

REFERENCES:

1. Network Intrusion Alert: An Ethical Hacking Guide to Intrusion Detection, Ankit Fadia, Manu Zacharia, Thomson Course Technology PTR, 2007
2. Ethical Hacking, Thomas Mathew, OSB Publisher, 2003
3. Hacking Exposed: Network Security Secrets & Solutions, Stuart McClure, Joel Scambray and George Kurtz, McGraw-Hill, 2005
4. Hands-On Ethical Hacking and Network Defense, Michael Simpson, Kent Backman, James Corley, Cengage Learning

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. Minor (CYS) VII Semester

(22MC1CY402) INCIDENT RESPONSE MANAGEMENT

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To gain knowledge on incident response and its regulations
- To get an insight into the stages of incident response
- To ensure the security of an organization by analyzing the history of attacks
- To assess the response through various metrics
- To explore incident response to the LIVE Scenario

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand how incidents are responded to at a high level

CO-2: Gain knowledge on breaches and various attacks

CO-3: Build an Incident response team to improve the security of an organization

CO-4: Evaluate the incident response through attacks

CO-5: Apply the Incident response to the threats in cloud and LIVE scenarios

UNIT-I:

Introduction: Computer Security Incident, Goals of Incident Response and its importance, GDPR and NIS regulations about incident response, The NIST five phases of Incident Response, Incident Response Process and Essential Components.

Stages of Incident Response: Incident definitions and severity criteria, Identification of threats and vulnerabilities, IR identification, IR Containment, Investigation, Eradication.

UNIT-II:

Evolution and Current Challenges: The history of data breaches, Modern cyber security evolution, preventing future breaches, Preparing for attacks, Assessing security safeguards, Aiding investigations and legal prosecutions, Bringing the organization together during crises, Ensuring the integration of security initiatives.

UNIT-III:

Organization of IR Team: Composition of IR team, Building of IR team, Reactive Security, Proactive Security, Operational Security, The Significance of three Security Pillars, Security operations and continuous monitoring: Captive SOC, Co-managed SOC, Fully managed SOC.

UNIT-IV:

Metrics for Incident Response: Key incident response metrics, Prevalence metrics, Effectiveness metrics, Time-based metrics, Understanding KPIs, Key metrics for a phishing attack, Incident response metrics in the cloud.

UNIT-V:

Incident Handling: The incident response process, Handling an incident, Handling an incident in a phishing scenario.

IR case studies: Data Breach/Hacking Incident, Live Ongoing Hacking Incident.

TEXT BOOKS:

1. Incident Response in the Age of Cloud: Techniques and Best Practices to Effectively Respond to Cybersecurity Incidents, Dr. Erdal Ozkaya, Packt Publishing, 2021
2. Incident Response & Computer Forensics, Jason T. Luttgens, Matthew Pepe, McGraw-Hill Education, 2014

REFERENCES:

1. Incident Response, Kenneth R. van Wyk and Richard Forno, O'Reilly Media, 2001
2. Intelligence-Driven Incident Response: outwitting the Adversary, Scott J. Roberts and Rebekah Brown, O'Reilly Media, 2017.

ONLINE RESOURCES:

1. Incident Response Lifecycle